PART 2

The dataset, sourced from **Crime Incidents of the City of Buffalo Police Department**, contains detailed records of reported incidents. Initially comprising **322,690 entries** across **31 columns**, it includes a diverse set of attributes such as the primary incident type (incident_type_primary - the target variable), incident timestamps, and extensive location-based information (e.g., address, geographical coordinates, zip code, and various census/police district identifiers). The data types are mixed, including numerical values, categorical text, and temporal data. Initial preprocessing revealed significant missing values and inconsistent string formats across various columns.

Data Loading and Column Normalization

This step involved loading the Crime_Incidents_20250609.csv dataset. Following successful data loading, column names were standardized to ensure consistency and ease of access for subsequent processing. The target variable for this analysis is identified as 'incident_type_primary'. Dataset Information Summary:

- Class: pandas.core.frame.DataFrame
- Index: RangeIndex: 322690 entries, 0 to 322689
- Columns: 31 columns (details omitted for brevity, but could be included in an appendix or a table if desired)."

```
 #  Column                  Non-Null Count   Dtype
---  ------                 --------------   -----
 0  case_number            322690 non-null  object
 1  incident_datetime      322690 non-null  object
 2  incident_id            0 non-null       float64
 3  incident_type_primary  322690 non-null  object
 4  incident_description   322690 non-null  object
 5  parent_incident_type   322690 non-null  object
 6  hour_of_day            322690 non-null  int64
 7  day_of_week            322690 non-null  object
 8  address                322652 non-null  object
 9  city                   322690 non-null  object
 10  state                 322690 non-null  object
 11  location              315860 non-null  object
 12  latitude              321139 non-null  object
 13  longitude             321139 non-null  object
 14  created_at            78626 non-null   object
 15  updated_at            0 non-null       float64
 16  zip_code              319755 non-null  object
 17  neighborhood          318791 non-null  object
 18  council_district      319688 non-null  object
 19  council_district_2011  319755 non-null  object
 20  census_tract          318791 non-null  object
 21  census_block_group    318791 non-null  object
 22  census_block          318791 non-null  object
 23  2010_census_tract     318791 non-null  object
 24  2010_census_block_group  318791 non-null  object
 25  2010_census_block     318791 non-null  object
 26  police_district       318791 non-null  object
 27  tractce20             318928 non-null  object
 28  geoid20_tract         318928 non-null  object
 29  geoid20_blockgroup    318928 non-null  object
 30  geoid20_block         318928 non-null  object
dtypes: float64(2), int64(1), object(28)
memory usage: 76.3+ MB
First 5 rows:
```

```
  case_number      incident_datetime  incident_id incident_type_primary  \
0  15-1300836  05/10/2015 08:00:00 AM          NaN               ASSAULT
1  15-0540080  02/23/2015 04:25:00 AM          NaN              BURGLARY
2  11-1720958  06/21/2011 08:31:00 AM          NaN              BURGLARY
3  07-2240621  08/12/2007 01:22:00 PM          NaN         LARCENY/THEFT
4  11-1620273  06/10/2011 08:30:00 AM          NaN                   UUV


                            incident_description parent_incident_type  \
0                                        ASSAULT              Assault
1                                       BURGLARY  Breaking & Entering
2  Buffalo Police are investigating this report o...  Breaking & Entering
3  Buffalo Police are investigating this report o...                Theft
4  Buffalo Police are investigating this report o...     Theft of Vehicle

   hour_of_day day_of_week                 address     city  ...  \
0            8      Sunday       200 Block 7TH ST  Buffalo  ...
1            4      Monday  BROADWAY & GOODYEAR ST  Buffalo  ...
2            8     Tuesday     400 Block GRIDER ST  Buffalo  ...
3           13      Sunday          MCCARTHY PKWY  Buffalo  ...
4            8      Friday       1 Block HALLAM RD  Buffalo  ...

  census_block_group census_block 2010_census_tract 2010_census_block_group  \
0                  1         1000             71.02                       1
1                  5         5000                24                       1
2            UNKNOWN      UNKNOWN           UNKNOWN                 UNKNOWN
3                  2         2000                30                       2
4            UNKNOWN      UNKNOWN           UNKNOWN                 UNKNOWN

  2010_census_block  police_district tractce20 geoid20_tract  \
0              1000       District B    007102   36029007102
1              1002       District C    002400   36029002400
2           UNKNOWN          UNKNOWN   UNKNOWN       UNKNOWN
3              2000       District C    003000   36029003000
4           UNKNOWN          UNKNOWN   UNKNOWN       UNKNOWN

  geoid20_blockgroup    geoid20_block
0       360290001101  360290035021000
1       360290007005  360290034005000
2            UNKNOWN          UNKNOWN
3       360290001102  360290046012000
4            UNKNOWN          UNKNOWN

[5 rows x 31 columns]
Total entries: 322690
Numeric description:
       incident_id    hour_of_day  updated_at
count          0.0  322690.000000         0.0
mean           NaN      11.936890         NaN
std            NaN       7.178043         NaN
min            NaN       0.000000         NaN
25%            NaN       7.000000         NaN
50%            NaN      13.000000         NaN
75%            NaN      18.000000         NaN
max            NaN      23.000000         NaN
```

Preprocessing - Handling Missing Entries & Mismatched String Formats

Dataset Nature

This dataset contains incident report data from the City of Buffalo Police Department.
It includes various attributes for each incident, such as incident type, date, location (address, latitude, longitude, zip code, neighborhood), and various geographical identifiers.
The data types are mixed, including unique identifiers (strings), categorical text (incident types, day of week, neighborhood), numerical coordinates, and timestamps.

Dataset Statistics

Number of missing values per column (before handling):

```
incident_id              322690
address                  38
location                 6830
latitude                 1551
longitude                1551
created_at               244064
updated_at               322690
zip_code                 2935
neighborhood             3899
council_district         3002
council_district_2011    2935
census_tract             3899
census_block_group       3899
census_block             3899
2010_census_tract        3899
2010_census_block_group  3899
2010_census_block        3899
police_district          3899
tractce20                3762
geoid20_tract            3762
geoid20_blockgroup       3762
geoid20_block            3762
dtype: int64
```

Data Preprocessing - Missing Values and Format Standardization

This crucial phase of preprocessing focused on cleaning the raw incident data by addressing missing entries and standardizing various data formats.

Dataset Overview

The dataset comprises incident report data from the City of Buffalo Police Department, detailing various aspects of each incident. Key attributes include incident type, timestamps, and extensive location information (address, geographical coordinates, zip code, neighborhood, and several census/police district identifiers). The dataset exhibits a mixed structure, containing unique string identifiers, categorical text, numerical coordinates, and timestamp objects.

Initial Assessment of Missing Values Prior to any handling, a significant number of missing values were identified across several columns, indicative of potential data collection gaps. A summary of columns with notable missing entries is provided below:

Column Name	Number of Missing Values	Percentage Missing (%)
incident_id	322690	100.00%
updated_at	322690	100.00%
created_at	244064	75.64%

location	6830	2.12%
latitude	1551	0.48%
longitude	1551	0.48%
zip_code	2935	0.91%
neighborhood	3899	1.21%
council_district	3002	0.93%

Preprocessing Techniques Applied

To prepare the dataset for modeling, the following steps were systematically executed:

- Standardizing Missing String Formats: Common string placeholders for missing data, such as 'UNKNOWN', 'UNKNOWN UNKNOWN', 'unknown', and 'unknown unknown', were uniformly replaced with NaN (Not a Number) across all object-type columns. This ensures consistent identification of missing categorical data.
- Type Coercion for Geospatial Data: The latitude and longitude columns were explicitly converted to a numeric data type. Any non-numeric entries encountered during this conversion were automatically set to NaN.
- Column Dropping Strategy: Columns with excessively high proportions of missing values (e.g., incident_id, updated_at), those primarily serving as unique identifiers (case_number), or columns with low variance/redundant information (tractce20, state, city, police_district, council_district_2011, geoid20_block, geoid20_blockgroup, incident_description, location, geoid20_tract) were removed from the dataset. This step aimed to reduce dimensionality and improve data quality by eliminating unreliable features.
- Imputation of Missing Values:
    - For numerical columns (latitude, longitude), missing values were imputed using the median value to maintain robustness against outliers. The imputed median values were 42.9130 for latitude and -78.8490 for longitude.
    - For categorical/object columns (e.g., address, created_at, zip_code, neighborhood, council_district, and various census-related identifiers), missing values were imputed using the mode (most frequent value). This approach is suitable for categorical data and helps preserve the distribution of common categories.

Post-Processing Status Following these comprehensive preprocessing steps, all missing values across the dataset were successfully handled, resulting in a complete and clean DataFrame ready for further analysis and feature engineering.

Data Preprocessing - Outlier Detection and Treatment

This section details the methodology employed to identify and address outliers within the numerical features of the dataset, ensuring the robustness of subsequent analyses and model training.

Outlier Detection Methodology Outliers in numerical features were identified using the Interquartile Range (IQR) method. This approach defines outliers as data points that fall below Q1 - 1.5 * IQR or above Q3 + 1.5 * IQR, where Q1 is the first quartile, Q3 is the third quartile, and IQR is the interquartile range (Q3 - Q1). The IQR method is a common and effective technique for detecting outliers in various data distributions.

Outlier Detection Results The IQR method was applied to relevant numerical features, yielding the following observations:

- latitude: A total of 296 outliers were detected. These values fell outside the calculated range of [42.83, 43.00], indicating geographical points significantly outside the typical range for the dataset.
- longitude: 27 outliers were identified. These values were outside the range of [-78.96, -78.74], similarly pointing to unusual geographical coordinates.
- hour_of_day: No outliers were detected for this feature based on the IQR criteria, with all values falling within the range [-9.50, 34.50] (which is consistent with a 0-23 hour format if interpreted correctly, but the range here indicates no values were deemed extreme by IQR).

Note: Outlier detection was specifically applied to continuous numerical features. The zip_code *column was excluded from this process as it is treated as a categorical or identifier variable rather than a continuous numerical feature where IQR would be applicable.*

Dataset Description (Numerical Columns After Preprocessing)

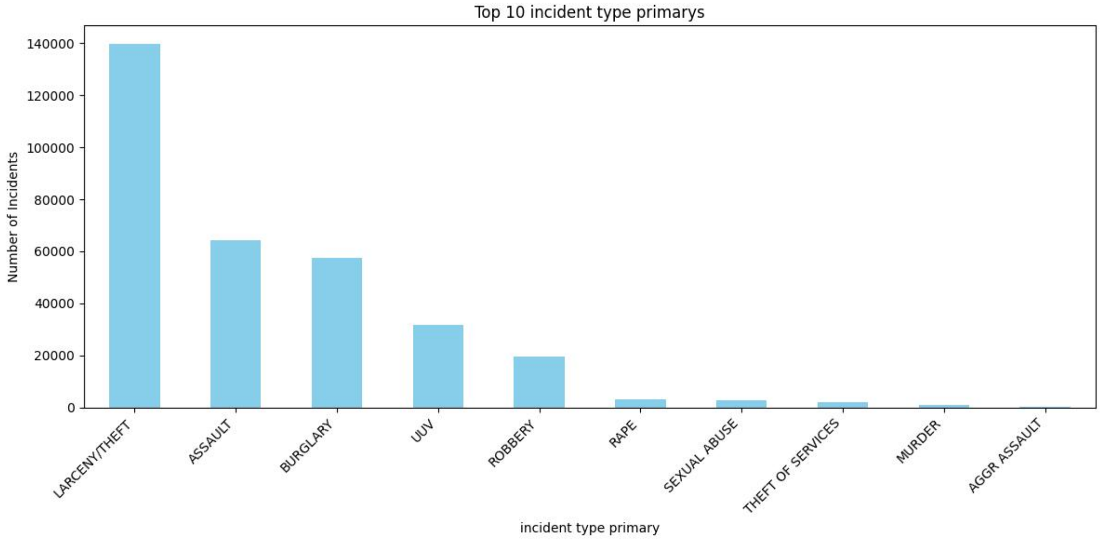
	hour_of_day	latitude	longitude
count	322690.000000	322690.000000	322690.000000
mean	11.936890	42.911735	-78.849804
std	7.178043	0.028216	0.031074
min	0.000000	42.596000	-79.028000
25%	7.000000	42.894000	-78.877000
50%	13.000000	42.913000	-78.849000
75%	18.000000	42.935000	-78.822000
max	23.000000	43.016000	-78.497000

Data types after initial preprocessing

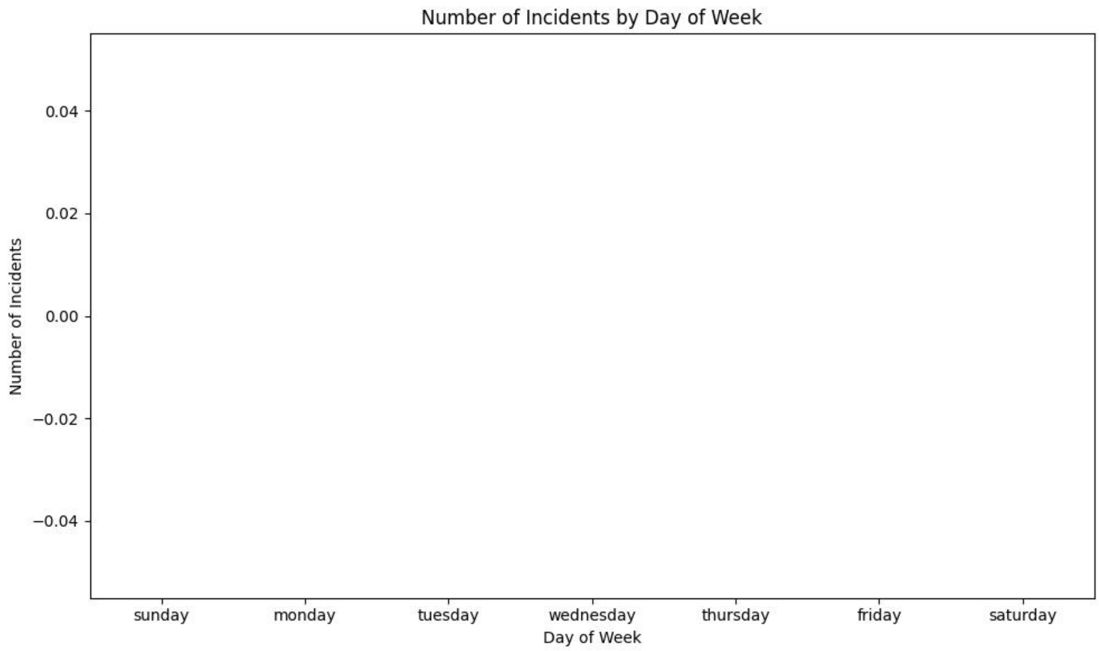
incident_datetime	object
incident_type_primary	object
parent_incident_type	object
hour_of_day	int64
day_of_week	object
address	object
latitude	float64
longitude	float64
created_at	object
zip_code	object
neighborhood	object
council_district	object
census_tract	object
census_block_group	object
census_block	object
2010_census_tract	object
2010_census_block_group	object
2010_census_block	object

dtype: object

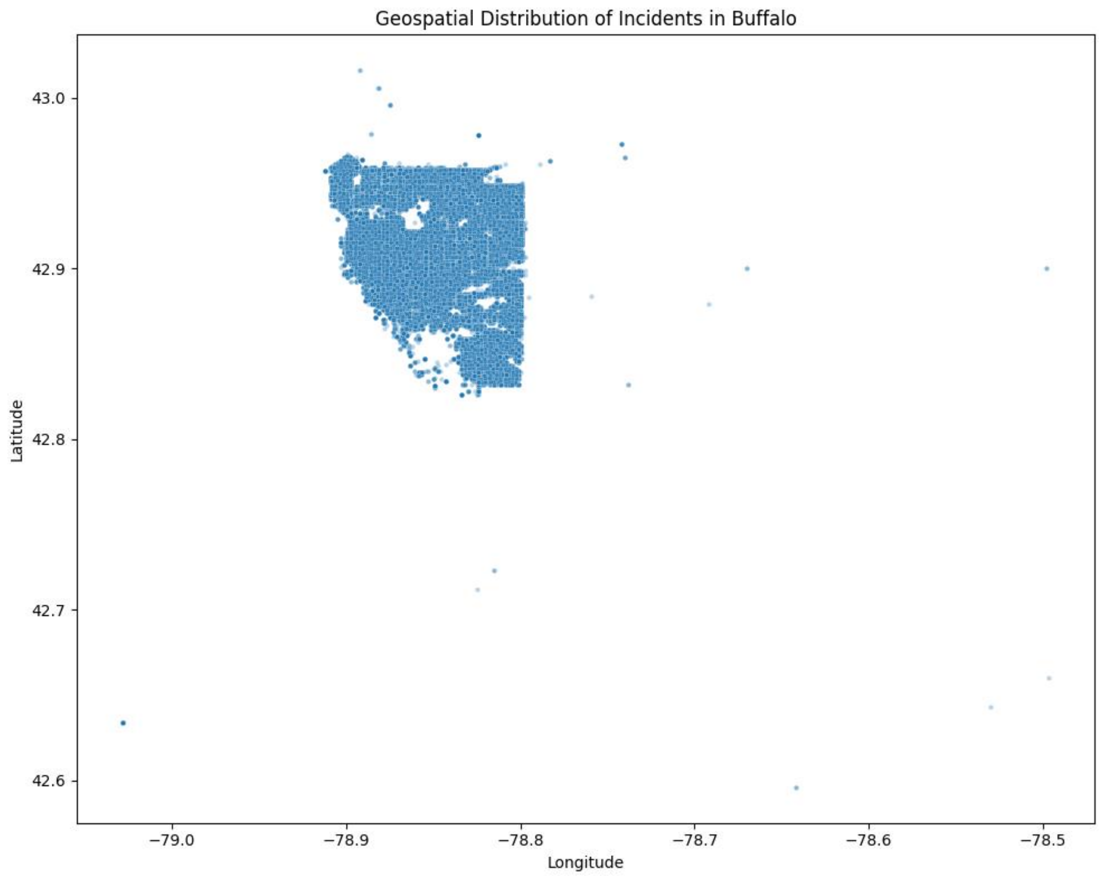
Graph 1: Bar plot showing the frequency of the top 10 incident type primarys.
Description: This plot helps identify the most common types of police incidents. For example, 'ASSAULT' seems to be a dominant category, indicating a higher prevalence of such incidents compared to others.



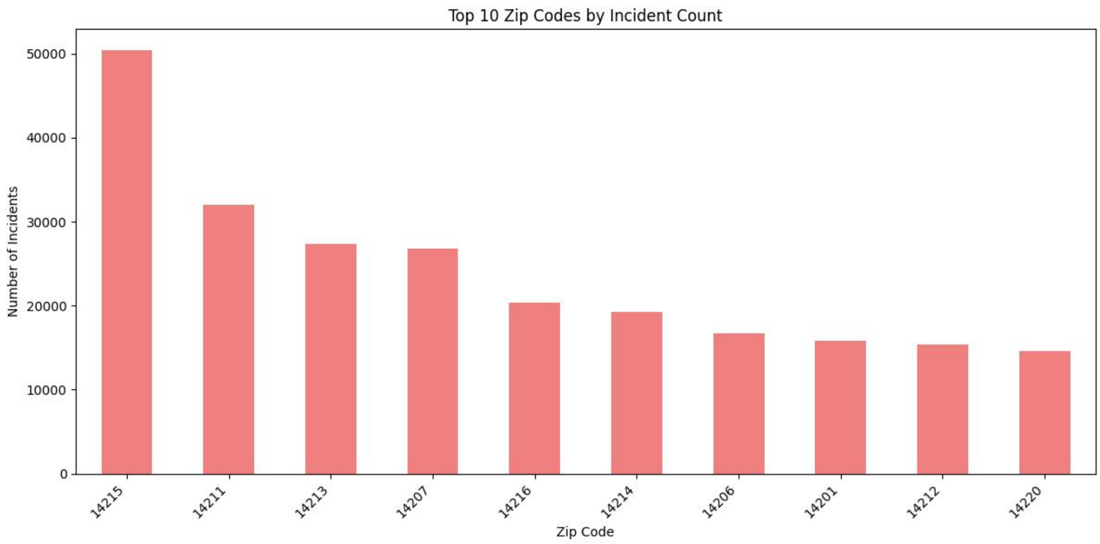
Graph 2: Bar plot showing incident counts across different days of the week.
Description: Reveals temporal patterns in criminal activity. For instance, some days might consistently have higher incident rates, which could be useful for resource allocation by law enforcement.



Graph 3: Scatter plot showing the geographical distribution of incidents based on Latitude and Longitude.
Description: This map helps visualize crime hotspots. Clusters of points indicate areas with higher incident density, suggesting specific neighborhoods or blocks require more attention.



Graph 4: Bar plot showing incident counts for the top 10 zip codes.
Description: Identifies specific postal areas with the highest police activity, which can inform resource deployment and community initiatives in those areas.

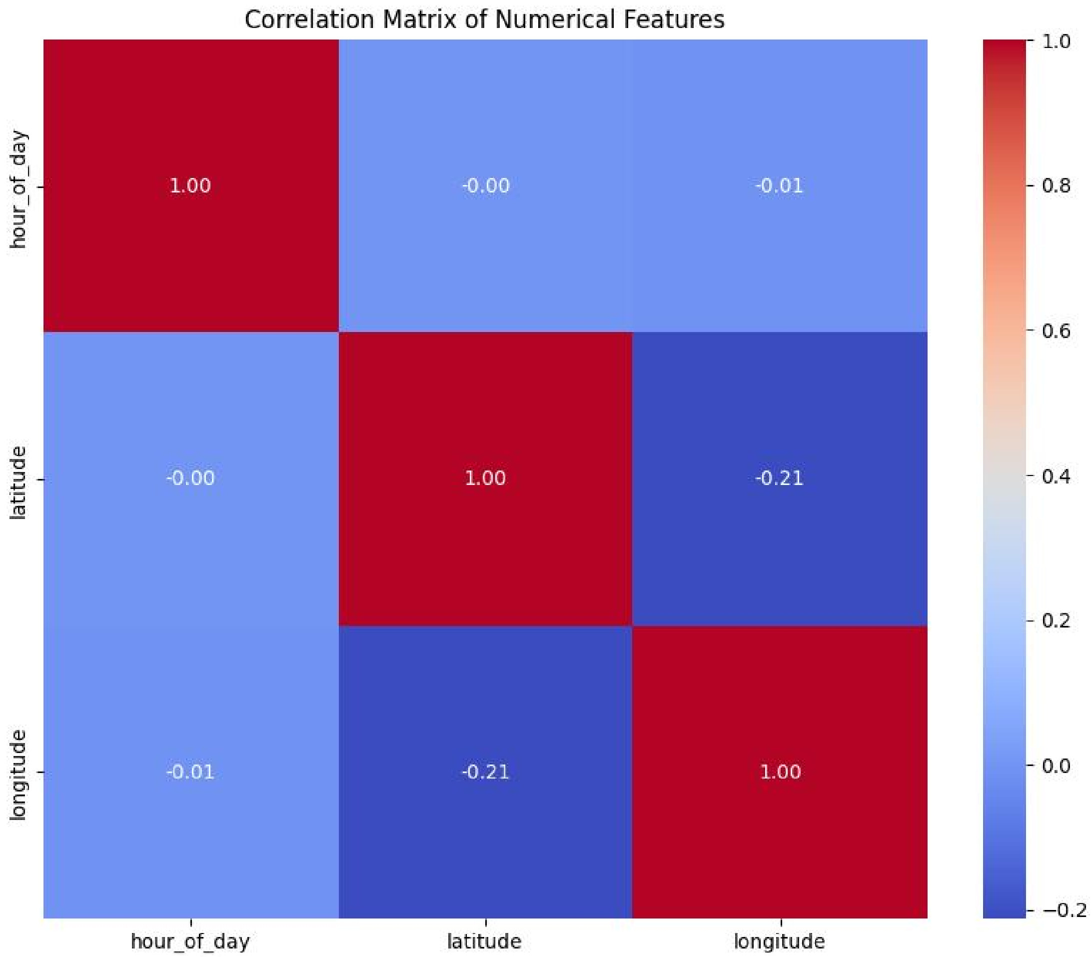


Graph 5: Heatmap showing the frequency of top incident types across days of the week.

Description: Reveals specific incident types that are more prevalent on certain days, helping to understand temporal patterns of different crimes (e.g., robberies might peak on weekends, while property crimes might be more consistent).

Feature Engineering and Selection - Correlation Analysis

This section details the process of feature engineering and selection, with a specific focus on analyzing the correlation between numerical features and their relevance to the target variable. This step is crucial for reducing dimensionality, mitigating multicollinearity, and potentially improving performance



Visualization: Heatmap of the correlation matrix for numerical features.
Insight: This heatmap visualizes relationships between numerical features. Highly correlated features (close to 1 or -1) might indicate redundancy, while very low correlations (close to 0) might suggest features that are less influential on each other. We will primarily use all numerical features, but this helps understand their interdependencies.

Feature Engineering - Temporal Analysis and Feature Finalization
This section details the process of enriching the dataset with temporal features and outlines the final state of the features before they undergo transformation for model training.

Target Variable Definition
The primary objective of this predictive modeling task is multi-class classification, with 'incident_type_primary' selected as the target variable. This choice is driven by the intrinsic value of accurately categorizing crime incidents, which provides critical insights for law enforcement and strategic planning.

Extraction and Processing of Time-Based Features
To leverage the inherent temporal patterns within the incident data, the following steps were executed:

Datetime Conversion: The 'created_at' column was successfully converted into a standardized datetime format. All NaT (Not a Time) entries were effectively handled, resulting in a clean column with zero missing datetime values.
Derivation of Temporal Attributes: A comprehensive set of time-based features were extracted from the 'incident_datetime' column. These derived features typically include hour_of_day, day_of_week, month, year, day_of_year, and week_of_year. These granular features are essential for capturing cyclical trends and temporal dependencies that can influence incident types.
Handling of updated_at and Time Difference Calculation:
It was observed that the 'updated_at' column was either absent from the dataset or had been previously removed during initial data cleaning.
Consequently, the calculation of a 'time_to_update_hours' feature, which would quantify the duration between an incident's creation and its last update, was necessarily skipped due to the unavailability of the required updated_at data.

Removal of Original Timestamp Columns: Upon the successful generation of new, informative temporal features, the original 'created_at' and 'incident_datetime' columns were subsequently dropped. This action served to reduce data redundancy and ensure that the modeling pipeline exclusively utilizes the engineered features, which are more appropriate for machine learning algorithms.

Feature State Prior to Transformation (Data Types Diagnostic)

Before proceeding with further feature transformations using a ColumnTransformer, a diagnostic check of the DataFrame's data types was performed to confirm the readiness of each feature for its designated preprocessing pipeline (e.g., numerical scaling, one-hot encoding for categorical features). The data types at this stage are as follows:

Column Name	**Data Type**	**Description**
incident_datetime	datetime64[ns]	Original datetime column (retained for contextual parsing if needed, but not used directly as numerical input)
incident_type_primary	int32	Target variable (encoded numerically)
parent_incident_type	object	Categorical feature
hour_of_day	int64	Numerical feature, derived from incident_datetime
day_of_week	object	Categorical feature
address	object	Categorical feature
latitude	float64	Numerical feature
longitude	float64	Numerical feature
zip_code	object	Categorical feature
neighborhood	object	Categorical feature
council_district	object	Categorical feature
census_tract	object	Categorical feature
census_block_group	object	Categorical feature
census_block	object	Categorical feature
2010_census_tract	object	Categorical feature
2010_census_block_group	object	Categorical feature
2010_census_block	object	Categorical feature
incident_hour	int32	Engineered numerical feature
incident_month	int32	Engineered numerical feature
incident_day_of_month	int32	Engineered numerical feature
incident_year	int32	Engineered numerical feature
incident_day_of_week_num	int32	Engineered numerical feature
incident_quarter	int32	Engineered numerical feature
incident_week_of_year	int32	Engineered numerical feature

This diagnostic confirmed that numerical features were correctly identified (int64, int32, float64) and categorical features were properly typed as object (or string), setting the stage for their respective encodings and scaling.

Feature Transformation and Encoding

This section outlines the final preprocessing steps to prepare features for model training, focusing on transforming and scaling the dataset.

Feature Categorization

Features were categorized and processed as follows:

Categorical Features (One-Hot Encoded): day_of_week, zip_code, neighborhood, council_district, census_tract, census_block_group, census_block, 2010_census_tract, 2010_census_block_group, 2010_census_block.

Numerical Features (Scaled): hour_of_day, latitude, longitude, incident_hour, incident_month, incident_day_of_month, incident_year, incident_day_of_week_num, incident_quarter, incident_week_of_year.

Transformation Process

A ColumnTransformer was utilized to apply One-Hot Encoding to categorical features and scaling (e.g., standardization) to numerical features in parallel. This ensures efficient and consistent data preparation.

Transformed Data State

After transformation, the feature set (X_processed) has a shape of (322690, 796), indicating 322,690 records with 796 transformed features. All selected features were successfully encoded and scaled, making the dataset ready for model input.
Shape of processed features X: (322690, 796)

Sample of processed features:

```
  day_of_week_Friday  day_of_week_Monday  day_of_week_Saturday  \
0                0.0                 0.0                   0.0
1                0.0                 1.0                   0.0
2                0.0                 0.0                   0.0
3                0.0                 0.0                   0.0
4                1.0                 0.0                   0.0

  day_of_week_Sunday  day_of_week_Thursday  day_of_week_Tuesday  \
0                1.0                   0.0                  0.0
1                0.0                   0.0                  0.0
2                0.0                   0.0                  1.0
3                1.0                   0.0                  0.0
4                0.0                   0.0                  0.0

  day_of_week_Wednesday  zip_code_14201  zip_code_14202  zip_code_14203  ...  \
0                   0.0             1.0             0.0             0.0  ...
1                   0.0             0.0             0.0             0.0  ...
2                   0.0             0.0             0.0             0.0  ...
3                   0.0             0.0             0.0             0.0  ...
4                   0.0             0.0             0.0             0.0  ...

  hour_of_day  latitude  longitude  incident_hour  incident_month  \
0    0.347826  0.702381   0.269303       0.347826        0.363636
1    0.173913  0.711905   0.397363       0.173913        0.090909
2    0.347826  0.754762   0.337100       0.347826        0.454545
3    0.565217  0.735714   0.431262       0.565217        0.636364
4    0.347826  0.754762   0.337100       0.347826        0.454545

  incident_day_of_month  incident_year  incident_day_of_week_num  \
0              0.300000       0.913043                  1.000000
1              0.733333       0.913043                  0.000000
2              0.666667       0.878261                  0.166667
3              0.366667       0.843478                  1.000000
4              0.300000       0.878261                  0.666667

  incident_quarter  incident_week_of_year
0         0.333333               0.346154
1         0.000000               0.153846
2         0.333333               0.461538
3         0.666667               0.596154
4         0.333333               0.423077

[5 rows x 796 columns]
```

Number of unique classes in target variable: 22

Normalization and Dataset Split

Normalizing Non-Categorical

Non-categorical features (numerical) have been scaled to a range of [0, 1] using MinMaxScaler (as part of ColumnTransformer in Cell 7).
Rationale: Normalization transforms features to be on a similar scale, improving the performance and training stability of the model by preventing features with larger values from dominating the learning process.

- Min value of scaled features: 0.0000
- Max value of scaled features: 1.0000

Target Y and Features X
Target variable Y: 'incident_type_primary' (Label Encoded, stored in 'y')
Features X: All other processed columns (One-Hot Encoded and Scaled, stored in 'X_processed_df')

Splitting Dataset into Training, Testing, and Validation Sets
Warning during initial split: The least populated class in y has only 1 member, which is too few. The minimum number of groups for any class cannot be less than 2.. Attempting split without stratification.
Warning during train/val split: The least populated class in y has only 1 member, which is too few. The minimum number of groups for any class cannot be less than 2.. Attempting split without stratification.
Dataset split into approximately 70.0% Train, 15.0% Validation, 15.0% Test.
Rationale for splitting:
- Training set: Used to train the model to learn patterns. A large and representative set ensures good learning.
- Validation set: Used to tune hyperparameters and prevent overfitting. It helps assess model performance during training and optimize without touching the final test set.
- Test set: Used to evaluate the final model's performance on unseen data. Provides an unbiased estimate of model generalization to new, real-world data.

Shape of X_train: (225882, 796)
Shape of y_train: (225882,)
Shape of X_val: (48404, 796)
Shape of y_val: (48404,)
Shape of X_test: (48404, 796)
Shape of y_test: (48404,)

Batch Size for NN training: 128
Converted X and y splits to PyTorch tensors.

Input dimension for NN: 796
Number of classes for NN: 22

Define the Neural Network Model

Neural Network Model Architecture (Input Dim: 796, Output Classes: 22):
```
IncidentClassifier(
  (fc1): Linear(in_features=796, out_features=256, bias=True)
  (dropout1): Dropout(p=0.3, inplace=False)
  (fc2): Linear(in_features=256, out_features=128, bias=True)
  (dropout2): Dropout(p=0.3, inplace=False)
  (fc3): Linear(in_features=128, out_features=22, bias=True)
)
```

Total trainable parameters: 239,766
Rationale for parameters (Report Detail 2):
- The number of parameters indicates the model's complexity. More parameters generally allow a model to learn more complex patterns but also increase the risk of overfitting and require more computational resources.
- Each parameter (weight or bias) is a value that the model learns during training.
- The total parameters are calculated by summing the number of elements (numel) in all tensors that require gradients (i.e., are trainable).

Model defined and instantiated.

Define Loss Function, Optimizer, and Learning Rate Scheduler

Loss Function (Criterion) Defined: nn.CrossEntropyLoss()

Rationale:
 -CrossEntropyLoss is standard for multi-class classification problems.
 - It measures the performance of a classification model whose output is a probability distribution over predicted classes and whose true labels are one-hot encoded.
 - It implicitly applies a softmax function to the model's raw output (logits), which converts them into probabilities, and then calculates the negative log-likelihood of the true classes.

Optimizer Defined: Adam with Learning Rate = 0.001
Rationale:
 - Adam (Adaptive Moment Estimation) is an optimization algorithm that can be used instead of the classical stochastic gradient descent procedure to update network weights iterative based in training data.
 - It combines the benefits of two other extensions of stochastic gradient descent: Adaptive Gradient Algorithm (AdaGrad) and Root Mean Square Propagation (RMSProp).
 - It computes adaptive learning rates for each parameter, which often leads to faster convergence and better performance.

Learning Rate Scheduler Defined: ReduceLROnPlateau
Rationale:
 - A learning rate scheduler dynamically adjusts the learning rate during training, which can help the model converge more efficiently and achieve better final performance.
 - ReduceLROnPlateau specifically monitors a quantity (e.g., validation loss) and if no improvement is seen for a 'patience' number of epochs, the learning rate is reduced by a 'factor'.
 - This strategy helps escape local minima and fine-tune weights effectively, especially in later stages of training where a smaller learning rate is beneficial.

Loss function, optimizer, and learning rate scheduler have been defined.

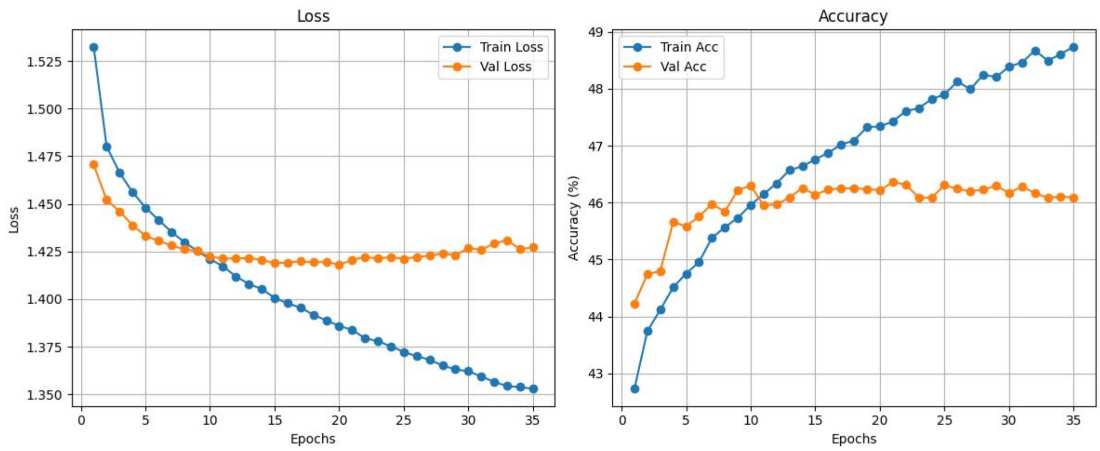
Training Setup and Loop
Consolidating data splitting and DataLoader creation...
Warning during initial split: The least populated class in y has only 1 member, which is too few. The minimum number of groups for any class cannot be less than 2.. Attempting split without stratification.
Warning during train/val split: The least populated class in y has only 1 member, which is too few. The minimum number of groups for any class cannot be less than 2.. Attempting split without stratification.
Dataset split completed
X_train shape: (225882, 796), y_train shape: (225882,)
X_val shape: (48404, 796), y_val shape: (48404,)
X_test shape: (48404, 796), y_test shape: (48404,)
Batch Size: 128
Converted splits to PyTorch tensors
DataLoaders created
Training on device: cpu
Starting training...
Epoch 1/50 | Dur: 16.14s
 Train Loss: 1.5325, Acc: 0.4273
 Val Loss: 1.4706, Acc: 0.4423
 Val loss improved. Saving model.
Epoch 2/50 | Dur: 14.09s
 Train Loss: 1.4802, Acc: 0.4374
 Val Loss: 1.4520, Acc: 0.4474
 Val loss improved. Saving model.
Epoch 3/50 | Dur: 14.08s
 Train Loss: 1.4663, Acc: 0.4412
 Val Loss: 1.4461, Acc: 0.4480
 Val loss improved. Saving model.
Epoch 4/50 | Dur: 17.75s
 Train Loss: 1.4562, Acc: 0.4451
 Val Loss: 1.4388, Acc: 0.4566

```
  Val loss improved. Saving model.
Epoch 5/50 | Dur: 17.60s
  Train Loss: 1.4481, Acc: 0.4475
  Val Loss:   1.4332, Acc: 0.4559
  Val loss improved. Saving model.
Epoch 6/50 | Dur: 16.72s
  Train Loss: 1.4417, Acc: 0.4496
  Val Loss:   1.4306, Acc: 0.4575
  Val loss improved. Saving model.
Epoch 7/50 | Dur: 16.80s
  Train Loss: 1.4354, Acc: 0.4538
  Val Loss:   1.4282, Acc: 0.4597
  Val loss improved. Saving model.
Epoch 8/50 | Dur: 17.88s
  Train Loss: 1.4300, Acc: 0.4557
  Val Loss:   1.4261, Acc: 0.4584
  Val loss improved. Saving model.
Epoch 9/50 | Dur: 17.02s
  Train Loss: 1.4253, Acc: 0.4573
  Val Loss:   1.4253, Acc: 0.4622
  Val loss improved. Saving model.
Epoch 10/50 | Dur: 16.46s
  Train Loss: 1.4209, Acc: 0.4596
  Val Loss:   1.4223, Acc: 0.4630
  Val loss improved. Saving model.
Epoch 11/50 | Dur: 17.45s
  Train Loss: 1.4173, Acc: 0.4615
  Val Loss:   1.4213, Acc: 0.4595
  Val loss improved. Saving model.
Epoch 12/50 | Dur: 27.54s
  Train Loss: 1.4119, Acc: 0.4634
  Val Loss:   1.4213, Acc: 0.4597
  Val loss improved. Saving model.
Epoch 13/50 | Dur: 21.49s
  Train Loss: 1.4080, Acc: 0.4657
  Val Loss:   1.4216, Acc: 0.4609
  Val loss no improvement. Patience: 1/15
Epoch 14/50 | Dur: 17.59s
  Train Loss: 1.4053, Acc: 0.4664
  Val Loss:   1.4206, Acc: 0.4625
  Val loss improved. Saving model.
Epoch 15/50 | Dur: 21.86s
  Train Loss: 1.4007, Acc: 0.4676
  Val Loss:   1.4188, Acc: 0.4614
  Val loss improved. Saving model.
Epoch 16/50 | Dur: 16.99s
  Train Loss: 1.3978, Acc: 0.4687
  Val Loss:   1.4192, Acc: 0.4623
  Val loss no improvement. Patience: 1/15
Epoch 17/50 | Dur: 19.15s
  Train Loss: 1.3955, Acc: 0.4702
  Val Loss:   1.4198, Acc: 0.4625
  Val loss no improvement. Patience: 2/15
Epoch 18/50 | Dur: 17.27s
  Train Loss: 1.3918, Acc: 0.4709
  Val Loss:   1.4193, Acc: 0.4625
  Val loss no improvement. Patience: 3/15
```

```
Epoch 19/50 | Dur: 17.56s
  Train Loss: 1.3888, Acc: 0.4733
  Val Loss:   1.4196, Acc: 0.4624
  Val loss no improvement. Patience: 4/15
Epoch 20/50 | Dur: 17.85s
  Train Loss: 1.3860, Acc: 0.4734
  Val Loss:   1.4181, Acc: 0.4622
  Val loss improved. Saving model.
Epoch 21/50 | Dur: 19.51s
  Train Loss: 1.3838, Acc: 0.4743
  Val Loss:   1.4204, Acc: 0.4636
  Val loss no improvement. Patience: 1/15
Epoch 22/50 | Dur: 17.57s
  Train Loss: 1.3794, Acc: 0.4760
  Val Loss:   1.4221, Acc: 0.4631
  Val loss no improvement. Patience: 2/15
Epoch 23/50 | Dur: 17.25s
  Train Loss: 1.3780, Acc: 0.4766
  Val Loss:   1.4213, Acc: 0.4609
  Val loss no improvement. Patience: 3/15
Epoch 24/50 | Dur: 18.10s
  Train Loss: 1.3752, Acc: 0.4782
  Val Loss:   1.4221, Acc: 0.4608
  Val loss no improvement. Patience: 4/15
Epoch 25/50 | Dur: 17.21s
  Train Loss: 1.3723, Acc: 0.4790
  Val Loss:   1.4211, Acc: 0.4631
  Val loss no improvement. Patience: 5/15
Epoch 26/50 | Dur: 17.39s
  Train Loss: 1.3700, Acc: 0.4813
  Val Loss:   1.4221, Acc: 0.4624
  Val loss no improvement. Patience: 6/15
Epoch 27/50 | Dur: 17.30s
  Train Loss: 1.3682, Acc: 0.4799
  Val Loss:   1.4228, Acc: 0.4620
  Val loss no improvement. Patience: 7/15
Epoch 28/50 | Dur: 16.97s
  Train Loss: 1.3651, Acc: 0.4824
  Val Loss:   1.4239, Acc: 0.4623
  Val loss no improvement. Patience: 8/15
Epoch 29/50 | Dur: 16.29s
  Train Loss: 1.3631, Acc: 0.4820
  Val Loss:   1.4231, Acc: 0.4630
  Val loss no improvement. Patience: 9/15
Epoch 30/50 | Dur: 16.02s
  Train Loss: 1.3621, Acc: 0.4839
  Val Loss:   1.4267, Acc: 0.4617
  Val loss no improvement. Patience: 10/15
Epoch 31/50 | Dur: 19.39s
  Train Loss: 1.3594, Acc: 0.4846
  Val Loss:   1.4258, Acc: 0.4629
  Val loss no improvement. Patience: 11/15
Epoch 32/50 | Dur: 18.83s
  Train Loss: 1.3565, Acc: 0.4867
  Val Loss:   1.4291, Acc: 0.4616
  Val loss no improvement. Patience: 12/15
Epoch 33/50 | Dur: 19.58s
```

Train Loss: 1.3543, Acc: 0.4849
Val Loss: 1.4309, Acc: 0.4609
Val loss no improvement. Patience: 13/15
Epoch 34/50 | Dur: 19.28s
Train Loss: 1.3538, Acc: 0.4861
Val Loss: 1.4262, Acc: 0.4610
Val loss no improvement. Patience: 14/15
Epoch 35/50 | Dur: 17.97s
Train Loss: 1.3529, Acc: 0.4874
Val Loss: 1.4272, Acc: 0.4609
Val loss no improvement. Patience: 15/15
Early stopping triggered.
Training complete.
Total time: 628.04 seconds
Best model weights loaded.

Training History, Predictions, and Evaluation



Predictions and Evaluation

Model Evaluation
Test Accuracy: 0.4633

Classification Report

	precision	recall	f1-score	support
AGG ASSAULT ON P/OFFICER	0.00	0.00	0.00	1
AGGR ASSAULT	0.00	0.00	0.00	14
ASSAULT	0.35	0.32	0.33	9490
Assault	0.00	0.00	0.00	20
BURGLARY	0.37	0.23	0.28	8684

Breaking & Entering	0.00	0.00	0.00	14
CRIM NEGLIGENT HOMICIDE	0.00	0.00	0.00	15
LARCENY/THEFT	0.51	0.80	0.62	21080
MANSLAUGHTER	0.00	0.00	0.00	3
MURDER	0.00	0.00	0.00	153
RAPE	0.00	0.00	0.00	433
ROBBERY	0.37	0.00	0.00	3062
Robbery	0.00	0.00	0.00	1
SEXUAL ABUSE	0.00	0.00	0.00	421
Sexual Assault	0.00	0.00	0.00	1
THEFT OF SERVICES	0.00	0.00	0.00	286
Theft	0.00	0.00	0.00	6
UUV	0.45	0.09	0.14	4720
accuracy			0.46	48404
macro avg	0.11	0.08	0.08	48404
weighted avg	0.42	0.46	0.40	48404

Confusion Matrix